GIS Programming and Customization
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LA Cultural Centers with Arcpy

Introduction.

The objective of this project is code script in Python to work with tabular data formats in a GIS, such as text and CSV files, then use this data to create new vector layers. Therefore, the data was created and contained the locations of Department of Cultural Affairs Cultural Centers and Theaters in Los Angeles. The goal is to show their geographic location within Los Angeles County, California by Python code script.

Study area.

The study area for this project is in Los Angeles, California, United States. Los Angeles is one of the largest cities located in Southern California. It is the center of the nation's film and television industry. Therefore a map of culture centers and theaters present high-quality arts exhibitions reflective of the diversity of Los Angeles.

	A	B	C	D
1	X_Long	Y_Lati	Z_Elevation_ft	CENTER_NAME
2	-118.187248	34.10422432	485.56	Art in the Park at Hermon Park
3	-118.294789	34.10169764	472.44	Barnsdall Art Center
4	-118.294789	34.10169764	472.44	Barnsdall Gallery Theatre
5	-118.294789	34.10169764	472.44	Barnsdall Junior Arts Center
6	-118.294789	34.10169764	472.44	Barnsdall Junior Arts Center Gallery
7	-118.2434695	34.05388374	587.27	Bridge Gallery at City Hall
8	-118.6001939	34.2014678	800.52	Canoga Park Youth Arts Center
9	-118.2146112	34.13928449	633.2	Center for the Arts Eagle Rock
10	-118.5024124	34.15965258	777.56	Encino Arts and Cultural Center
11	-118.2945679	34.10169824	6.56	Hollyhock House
12	-118.373542	34.16325736	1131.89	Lankershim Arts Center

Data and Data Processing.

The dataset contains 29 rows of data points and 4 columns of variables in this project. The column names are center name, x_long, y_lati, z_elevatuin that represent the name and the geographic coordinates of DCA Cultural Centers and Theaters in Los Angeles. Also, the CSV file was initially created without a projection. After being converted into a shapefile, the projection is WGS 1984.

Methods.

This process was completed by obtaining the necessary datasets from the respective project 2 folder provided from H drive. After examining the python files that had geoprocessing scripts, I picked 29 location points of DCA cultural centers and theaters in Los Angeles provided by the department of cultural affairs, then collected their geographic coordinates from Google Earth and imported this information into the excel sheet as a CSV file.

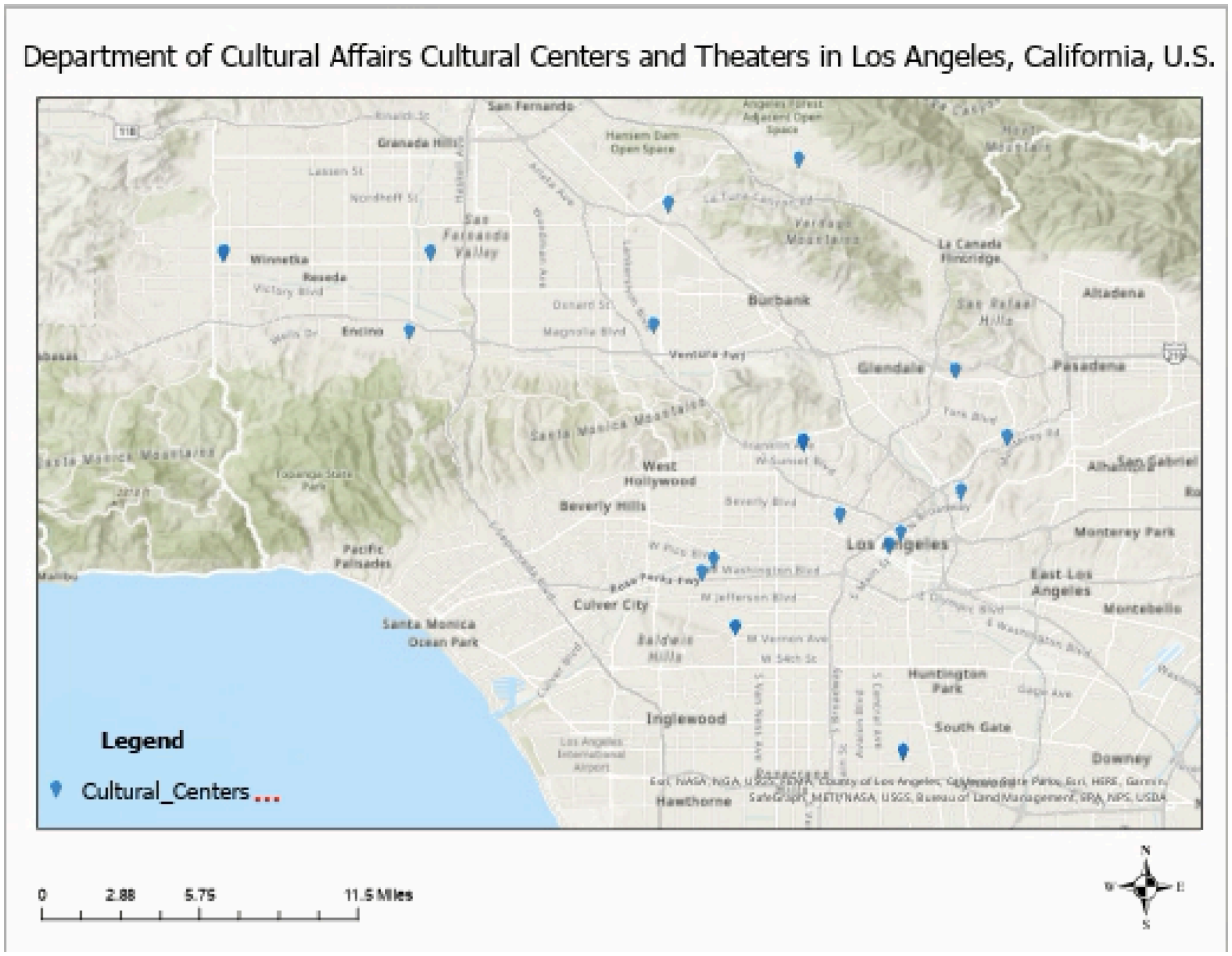
According to the provided python script template from the example, I modified the environment settings and local variables based on the created dataset. The x, y, and z coordinates of this script match the column names in the CSV file, and the path of files are changed as well. After the scripts successfully created the shapefiles, the Python Shell came out with the results 29 which exactly equal to the number of my locations. And the results come with the data points visualized on the map in ArcGIS Pro.

```
# import system modules
import arcpy

# Set environment settings
arcpy.env.workspace = "G:/586/project2"

# Set the local variables
in_table = "G:/586/project2/DCA_Cultural_Centers_and_Theaters.csv"
out_feature_class = "DCA_Cultural_Centers_and_Theaters.shp"
x_coords = "X_Long"
y_coords = "Y_Lati"
z_coords = "Z_Elevation_ft"

# Make the XY event layer...Source: https://spatialreference.org/ref/epsg/4326/,
arcpy.management.XYTableToPoint(in_table, out_feature_class,
                                x_coords, y_coords, z_coords,
                                arcpy.SpatialReference(4326))
# Print the total rows
print(arcpy.GetCount_management(out_feature_class))
```

Results.

Based on the methods and code screenshot, we can see the XYTable function at Arcpy module convert the CSV files into shapefiles.Then the GetCount_management and print functions

located in the last line get the total rows of the dataset and print the results which is 29. After checking the generated shapefiles, it shows the location points correctly in the ArcGISPro. Therefore, we can conclude this project helps us understand how to use Python in GIS to explore geographic coordinates in spatial data and practice the application of using ArcPy to modify spatial and tabular data then convert and visualize these to a vector data format.



Discussion.

The results of this project generate the finished map below containing the locations and distribution of DCA Culture Centers and Theaters. These centers and theaters offer high-quality instruction in the arts; produce solo and group art exhibitions; create outreach programs for underserved populations; develop special initiatives for young people, and promote numerous events during the year that celebrate the cultural diversity of the community. And we can see the data points seem evenly distributed in the Los Angeles area and this map clearly shows the centers and theaters locations which means people can choose the nearest place to visit based on this map.

However, the shapefiles and dataset only contains the name and geographic coordinates of the locations, the limitation is people need to search further information from the map for the actual address. For spatial data science discipline, this python script can identify the properties distribution under real estate management and construction companies such as tripalink near school. This can help companies evaluate the correlation between the locations of properties and the popularity of properties(lease amount, lease price, tenants turnover, etc.)

References.

https://culturela.org/cultural-centers/

https://earth.google.com/web/